

1.3:

Let E be an unsorted set of n segments that are the edges of a convex polygon. Describe an $O(n \log n)$ algorithm that computes from E a list containing all vertices of the polygon, sorted in clockwise order

Create an array of endpoints of the segments in an array A

For each segment $s = (u, v) \in E$, add u and v to A so that $|A| = 2n$

Sort the segments in A by their x coordinates, and if the x coordinates match for two segments, then by their y coordinates.

Scan the sorted array of vertices and remove duplicate vertices so all vertices are unique. This means that $|V| = n$ (# of segments)

Pick the vertex in V that has the lowest y value (if tied with another, then lowest x as well)

Indicate this with p_0

Sort all of the other vertices by the polar angle around p_0 in $O(n \log n)$

For each $p_i \in V \setminus \{p_0\}$, sort by the polar angle of vector $p_0 p_i$ by comparing the two points p_i and p_j using the cross product.

$p_i < p_j$ if $(p_0 p_i) \times (p_0 p_j) > 0$

If $x = 0$, break the tie by distance from p_0

This gives the points in counterclockwise order around p_0

Now the list is sorted in counterclockwise order $= [p_0, p_1, p_2, \dots, p_{n-1}]$ But we need clockwise order, so output the list in reverse order besides p_0

This would equal $[p_0, p_{n-1}, p_{n-2}, \dots, p_1]$

Then, it is in the indicated sorted order.

1.9:

Suppose that we have a subroutine CONVEXHULL available for computing the convex hull of a set of points in the plane. Its output is a list of convex hull vertices, sorted in clockwise order. Now let $S = \{x_1, x_2, \dots, x_n\}$ be a set of n numbers. Show that S can be sorted in $O(n)$ time plus the time needed for one call to CONVEXHULL. Since the sorting problem 16 has an $\Omega(n \log n)$ lower bound, this implies that the convex hull problem Section 1.5 EXERCISES has an $\Omega(n \log n)$ lower bound as well. Hence, the algorithm presented in this chapter is asymptotically optimal.

Use convex hull function CONVEXHULL(P) by encoding numbers to points in convex position so that the clockwise hull order can show the sorted order.

Use convex function $f(x) = x^2$ so that for any points on the parabola, the connecting segment lies above the curve. Also all other points on lower hull are in ascending order.

Given $S = \{x_1, \dots, x_n\}$ build a point set $P = \{p_i = (x_i, x_i^2) \mid i = 1, \dots, n\}$

Create points $p_i = (x_i, x_i^2)$ for all i .

Time = $O(n)$

Call $L \leftarrow \text{CONVEXHULL}(P)$ which will list vertices in clockwise order

Find the min and max points on the hull. Scan L once.

$p_{\min}$ = vertex in L with the smallest x

$p_{\max}$ = vertex in L with the largest x

Time = $O(n)$

Get the lower chain

Go from $p_{\min}$ to $p_{\max}$ in clockwise direction and count the amount of steps

If it is adjacent (just 1 step), then that direction is the upper edge, so walk the other way around the cycle to go from $p_{\min}$ to $p_{\max}$

Intake the x -coordinates along the direction that visits all direct vertices.

These x 's are in increasing order $\rightarrow x_1 < x_2 < x_n$

This results in getting the set of n numbers in ascending order, and is therefore sorted.

Total time = $O(n)$

Sorry if the latex document is messy. This is my first time using it.